

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1212

Hartland 345 kV
April 26, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1212. The Interconnection Customer proposes a 332.7 MW battery energy storage facility interconnecting at the Hartland 345 kV substation in the Kentucky area. The request was received on May 23, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 481 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,395,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Westne Sw Yd to Poplar Bluff Generating Sta 138kV	Hubbardstown to Tristate 345kV	145.6	280	5.5
Poplar Bluff Generating Station 345-138 kV Auto	Hubbardstown Substation to Tristate 345kV Line	139.2	418	8.6
Mcloughlin to Stateline 138kV	Hubbardstown to Tristate Substation 345kV	120.1	322	6.6
Hartland 230 kV/1 kV Transformer Bank	Tristate to Hubbardstown 345kV	108.2	270	6.9
Hubbardstown to Tristate 345kV	International Paper to Hanging Rock 500kV	107.8	1,562	32.8
Valemount 138/1kV Transformer	Pence to Valemount 345kV L	104.2	543	7.0
Stateline to Chadwick Substation 138kV L	Hubbardstown to Tristate 345kV	100.9	275	5.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Westone Sw Yd to Poplar Bluff Generating Sta Substation 138kV line	Rebuild line with higher-capacity conductor	\$13,208,000	14% (\$1,880,000)
Poplar Bluff Generating Sta 345/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$15,670,000	13% (\$1,997,000)
Mcloughlin to Stateline 138kV	Rebuild line with higher-capacity conductor	\$1,706,000	20% (\$332,000)
Hartland 230 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$6,388,000	25% (\$1,596,000)
Hubbardstown to Tristae 345kV	Reconductor line	\$8,841,000	9% (\$811,000)
Valemount 138 kV/1 kV TX	Replace transformer with higher-rated unit	\$4,679,000	11% (\$498,000)
Stateline to Chadwick Substation 138kV L	Reconductor line	\$1,234,000	23% (\$281,000)
	Total allocated to project		\$7,395,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.